

Tunnel Diodes, RTDs and Tunnel Transistors

KKKC6353 Nanoelectronic Devices · Lecture 3, Part 2 of 3

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Three ways to use tunnelling

Tunnel diode

A p–n junction so heavily doped that its wall is about 10 nm thick. Its current rises, falls, then rises again.

Resonant-tunnelling diode

Two thin walls around a quantum well: a sharp energy filter, and the fastest electronic oscillator at room temperature.

Tunnel FET

A transistor that switches by opening a tunnelling window, to beat the 60 mV limit of ordinary transistors.

1

The tunnel diode

Make the wall of a p–n junction thin enough to tunnel through.

A thin wall in a p–n junction

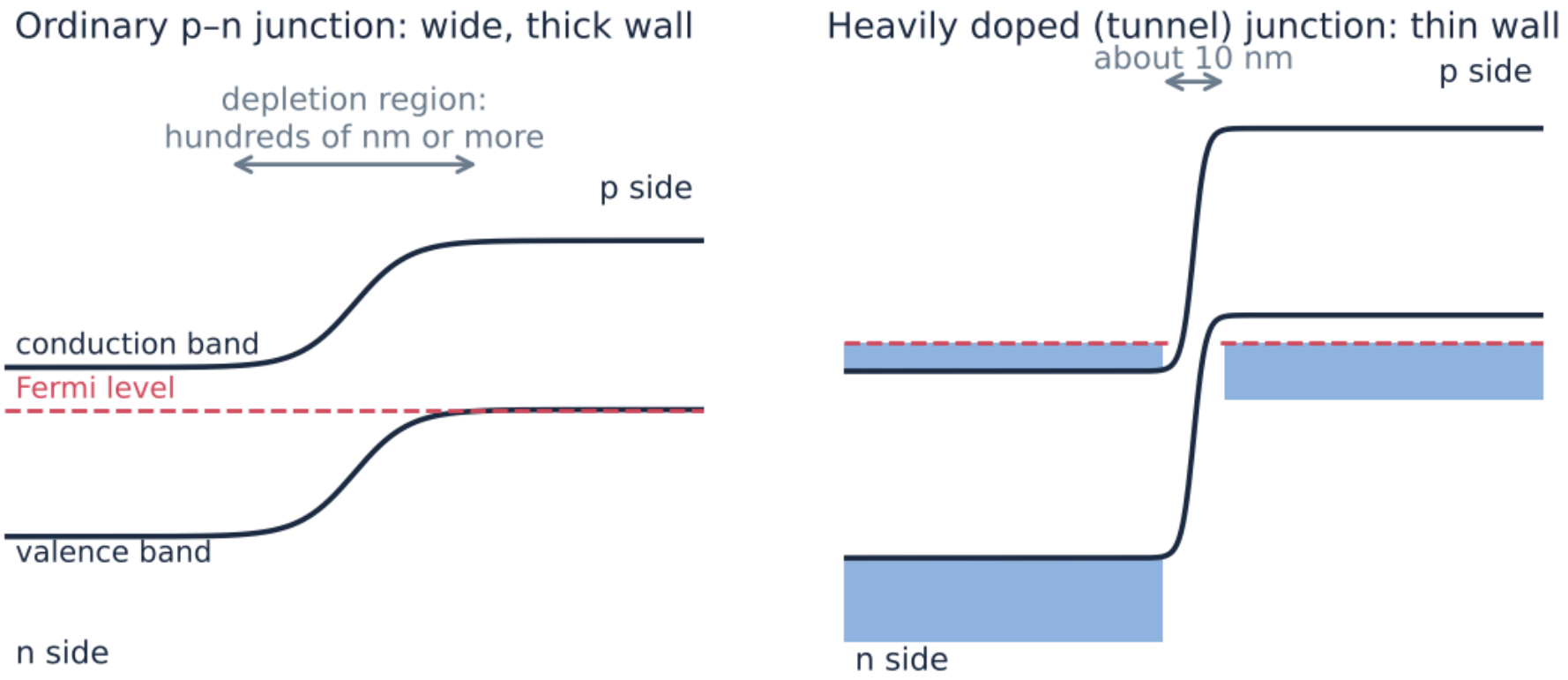


Figure made for this course; schematic.

Dope both sides about **a thousand times** more heavily (10^{19}–10^{20} cm^{-3}): the depletion region shrinks to **about 10 nm**, and the Fermi level moves **into** the bands.

Esaki's discovery

1957, Tokyo

Leo Esaki, at the company soon renamed Sony, studies very heavily doped germanium junctions.

Something new

As the forward voltage rises, the current rises, then falls, then rises again. He explains it as tunnelling (Physical Review, 1958).

Nobel Prize 1973

Shared with Ivar Giaever and Brian Josephson, both for tunnelling in superconductors. Josephson returns in Part 3.

2

Walking along the curve

Filled states must face empty states.

Step by step

- **Rule:** an electron keeps its energy, so it needs an **empty state at the same energy** opposite
- **(b)** No voltage: no net current
- **(c)** Filled strip faces empty strip: **peak**
- **(d)** The strips slide past: filled states face the gap, current **falls**
- **(e)** Electrons go over the step: an ordinary diode
- **(a)** Reverse: a large tunnelling current at once

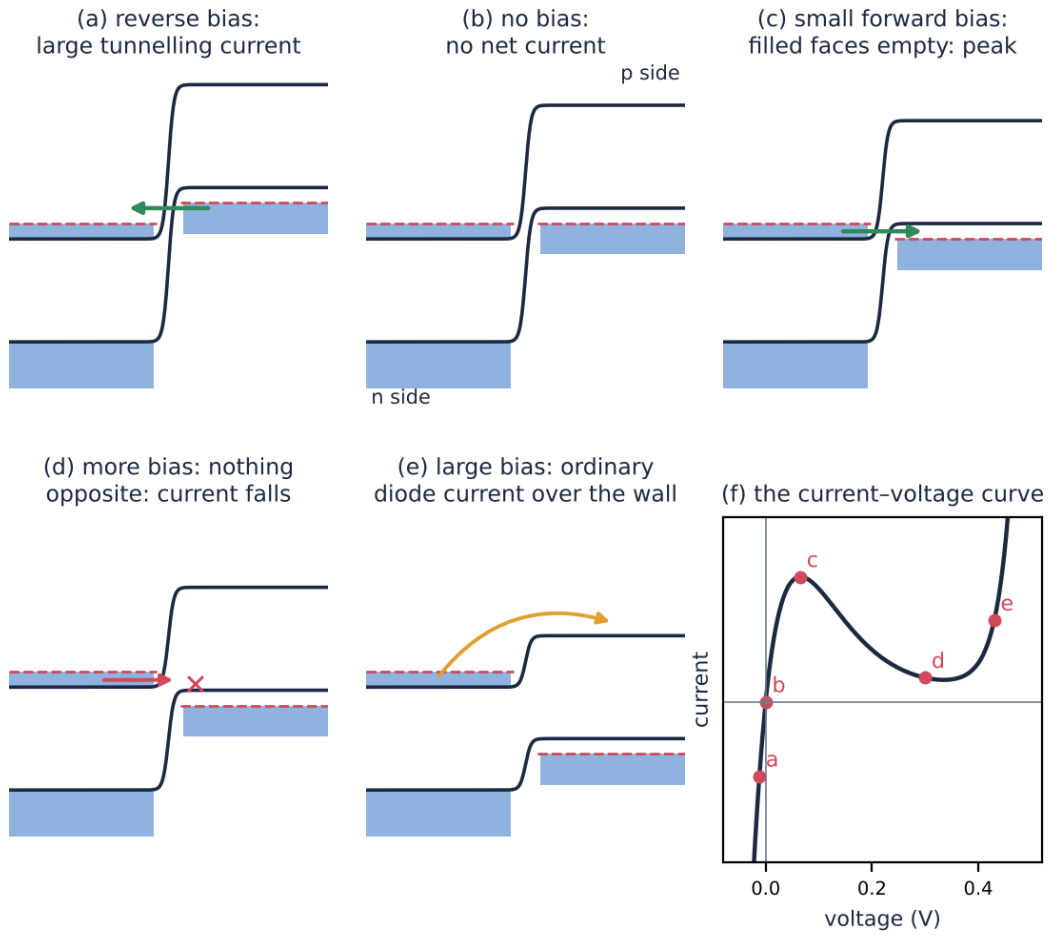


Figure made for this course; schematic.

Three currents add up

- **Tunnelling** makes the first hump
- The **ordinary diode** current makes the final rise
- A small **excess current** through defect states fills the valley
- Figure of merit: the **peak-to-valley current ratio** (about 5.6 here)
- Germanium: peak at a few tens of mV, valley at a few hundred mV

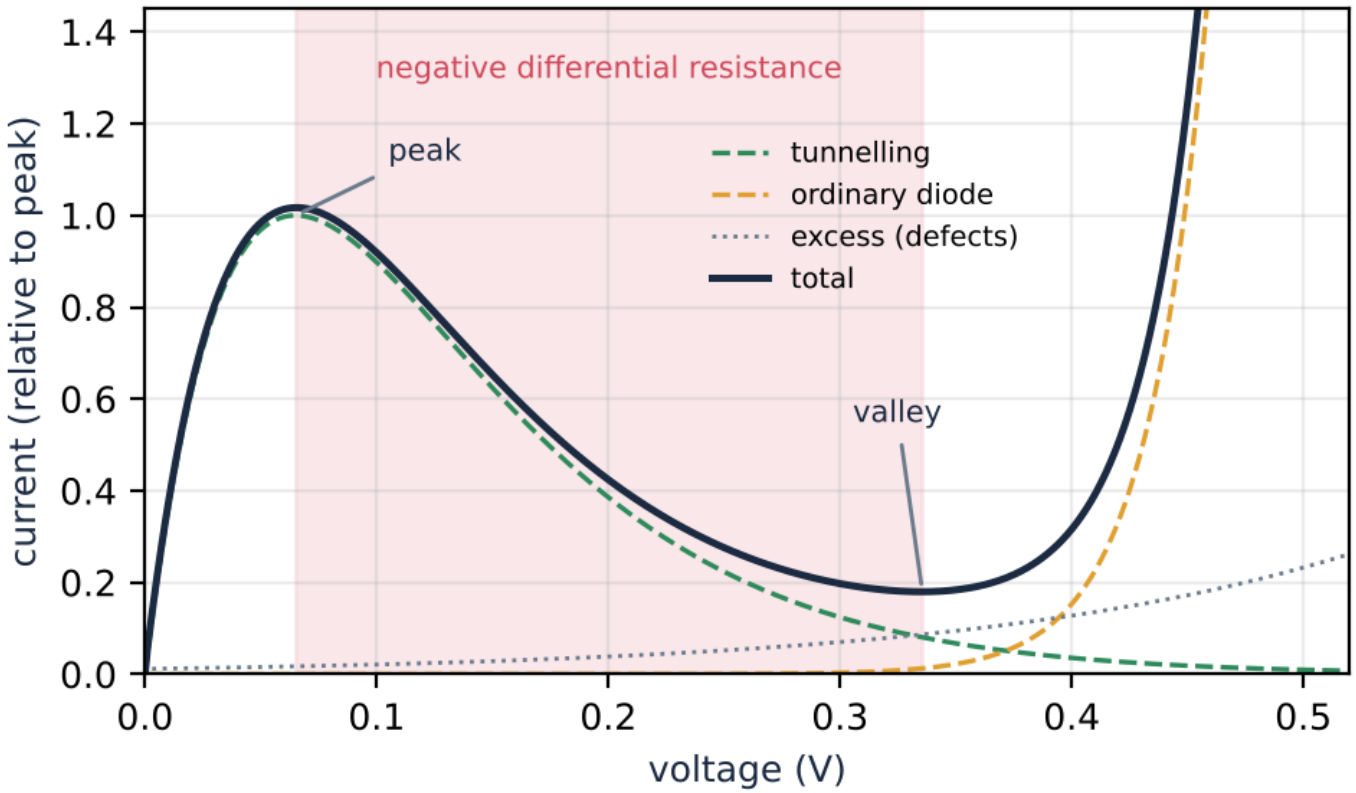


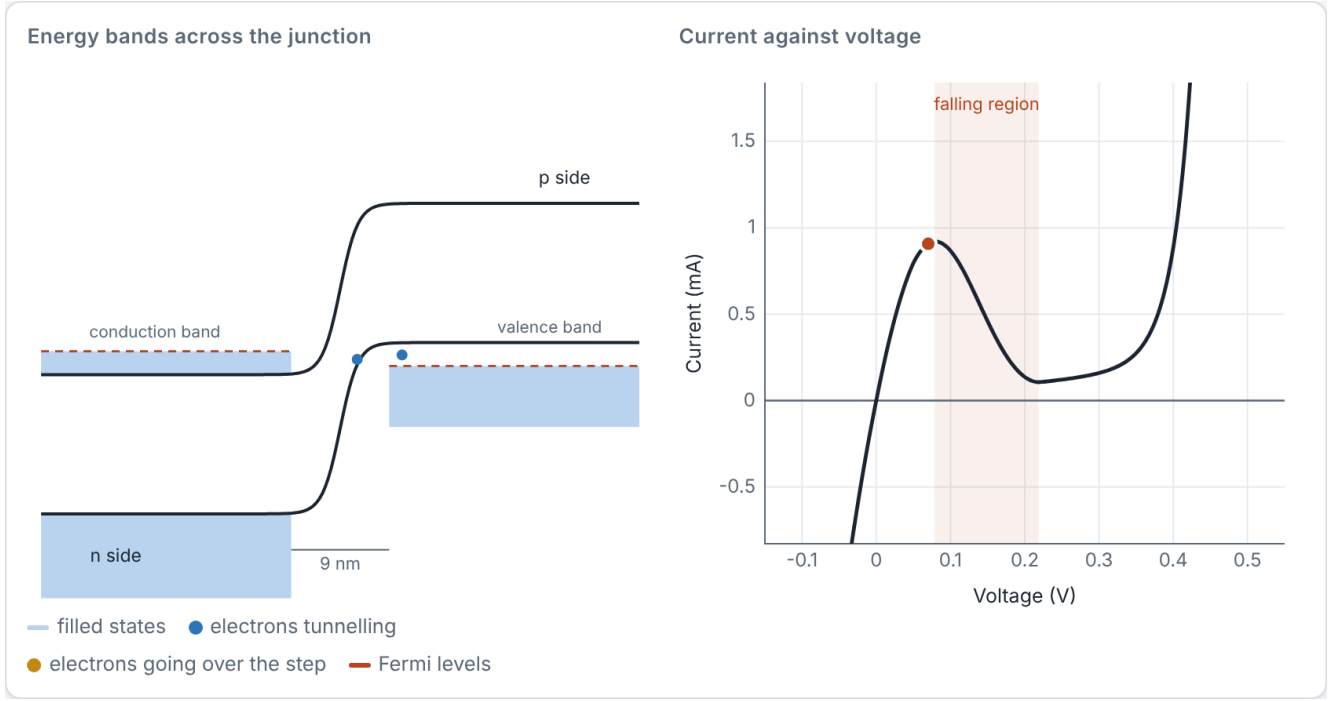
Figure made for this course from the model in the notes.

Interactive animation 3

Tunnel diode

faizverse.github.io/nanoelectronic-devices/L3/tunnel-diode.html

- Raise the voltage slowly and stop at the peak: what faces what?
- Go on to the valley: what lies opposite the filled strip now?
- Lower the doping: what happens to the peak, and why?
- Switch to gallium arsenide: when does the ordinary current start?



3

What a falling current is good for

Negative differential resistance makes oscillators.

Negative differential resistance

What it means

Between peak and valley, more voltage gives less current. The change in voltage divided by the change in current is negative.

A common misconception

“The diode produces power.” It does not: current and voltage are both positive, so it still absorbs power. Only the changes have opposite signs.

Oscillators

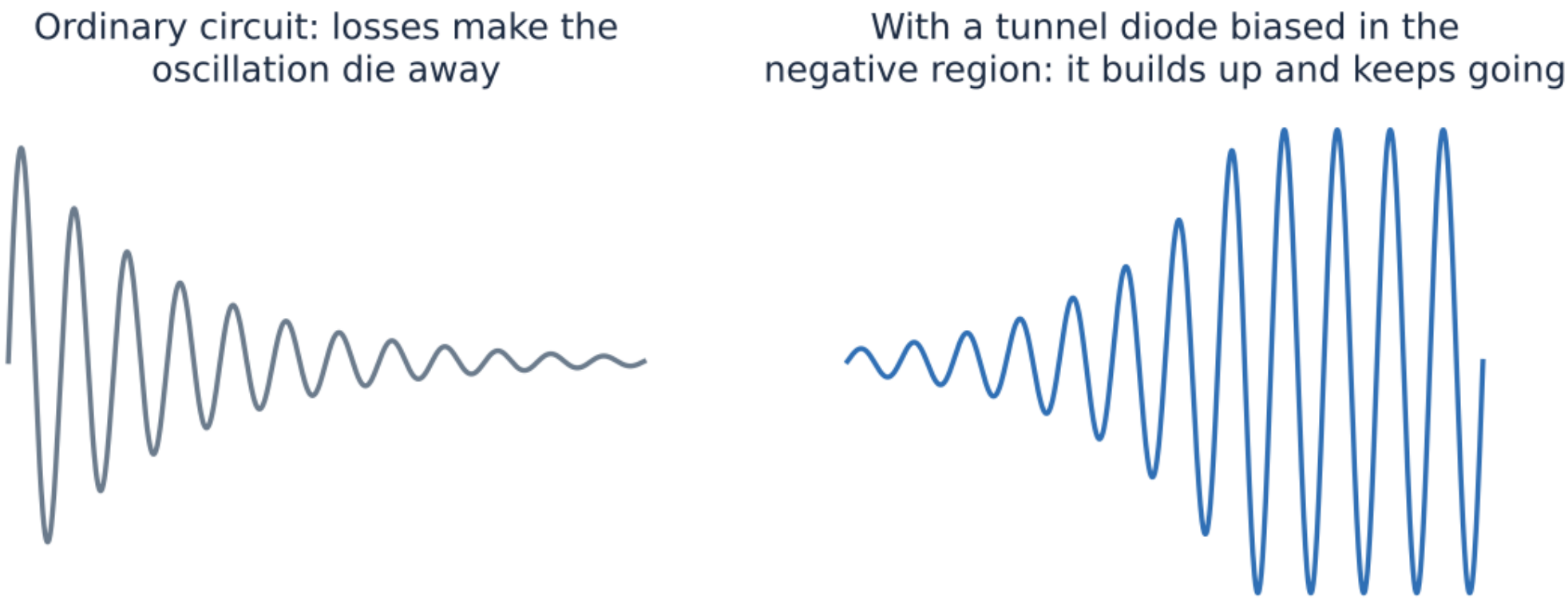


Figure made for this course; schematic.

A resonant circuit rings and dies away because of its losses. A tunnel diode biased in its falling region **cancels the losses**, so a tiny disturbance grows into a steady oscillation.

Where tunnel diodes went

- **Two terminals**: input and output share one pair of wires
- Small current; a narrow falling region
- Transistors could be packed by the **billion**; tunnel diodes could not
- Still at work: **tunnel junctions** join the cells of the best solar cells, on almost every satellite
- Relatives, **backward diodes**, detect microwaves

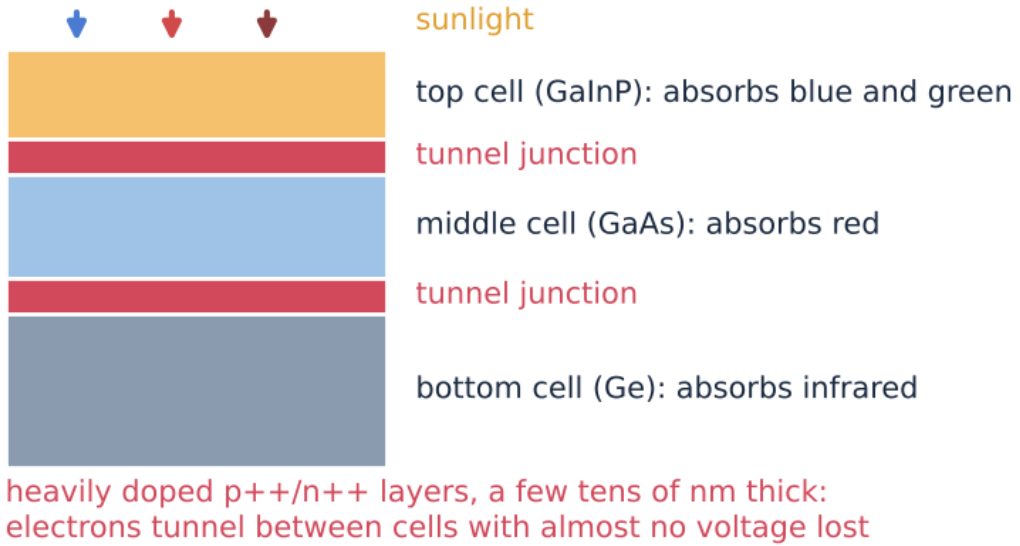


Figure made for this course; layer thicknesses not to scale.

4

The resonant-tunnelling diode

Two walls make a filter.

Two thin walls with a quantum well between

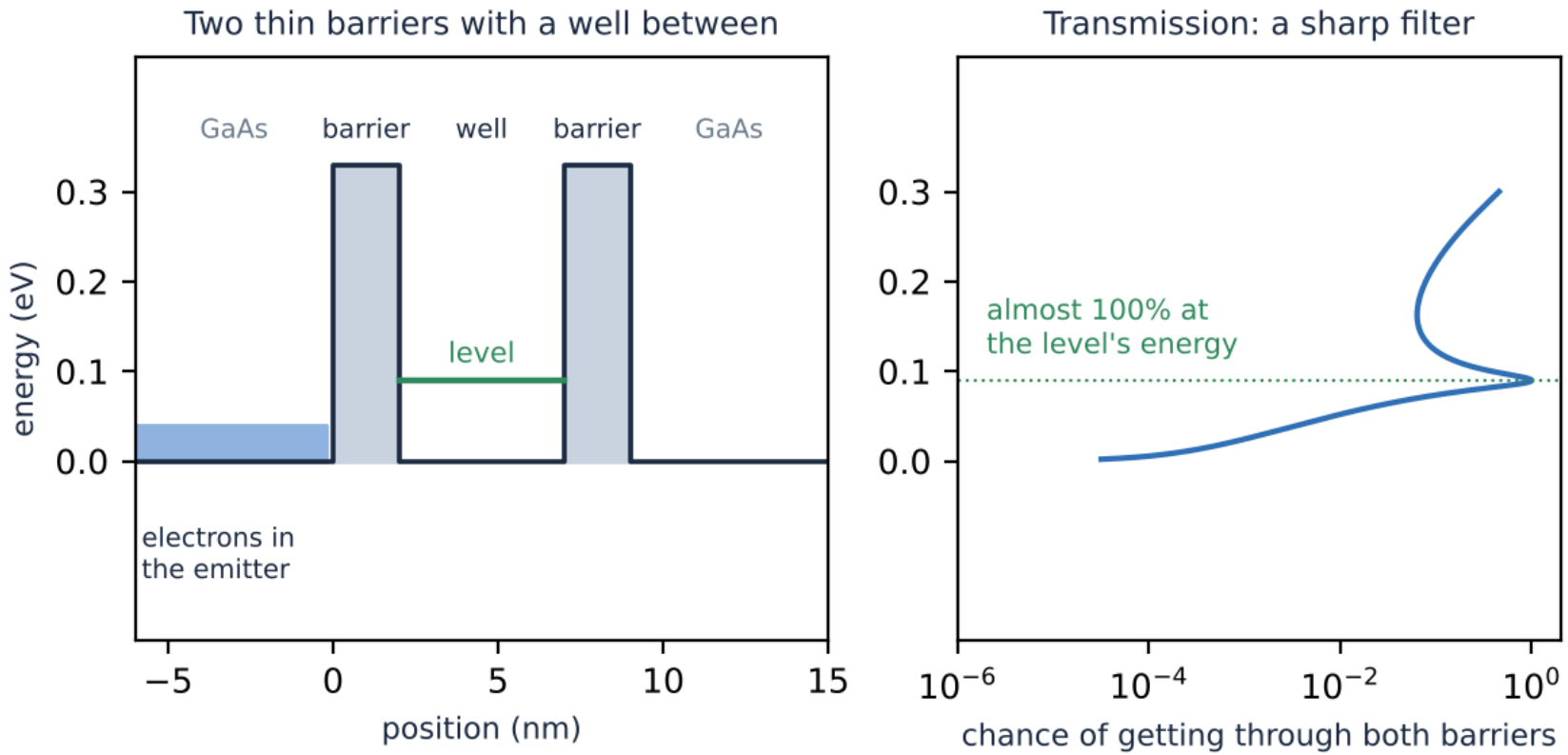


Figure made for this course, from an exact calculation (effective mass 0.067 m_0).

At the well's level the reflections from the two walls **cancel**, and the electron passes through both almost every time. At other energies it is almost always reflected: a **sharp filter**.

Moving the level with the voltage

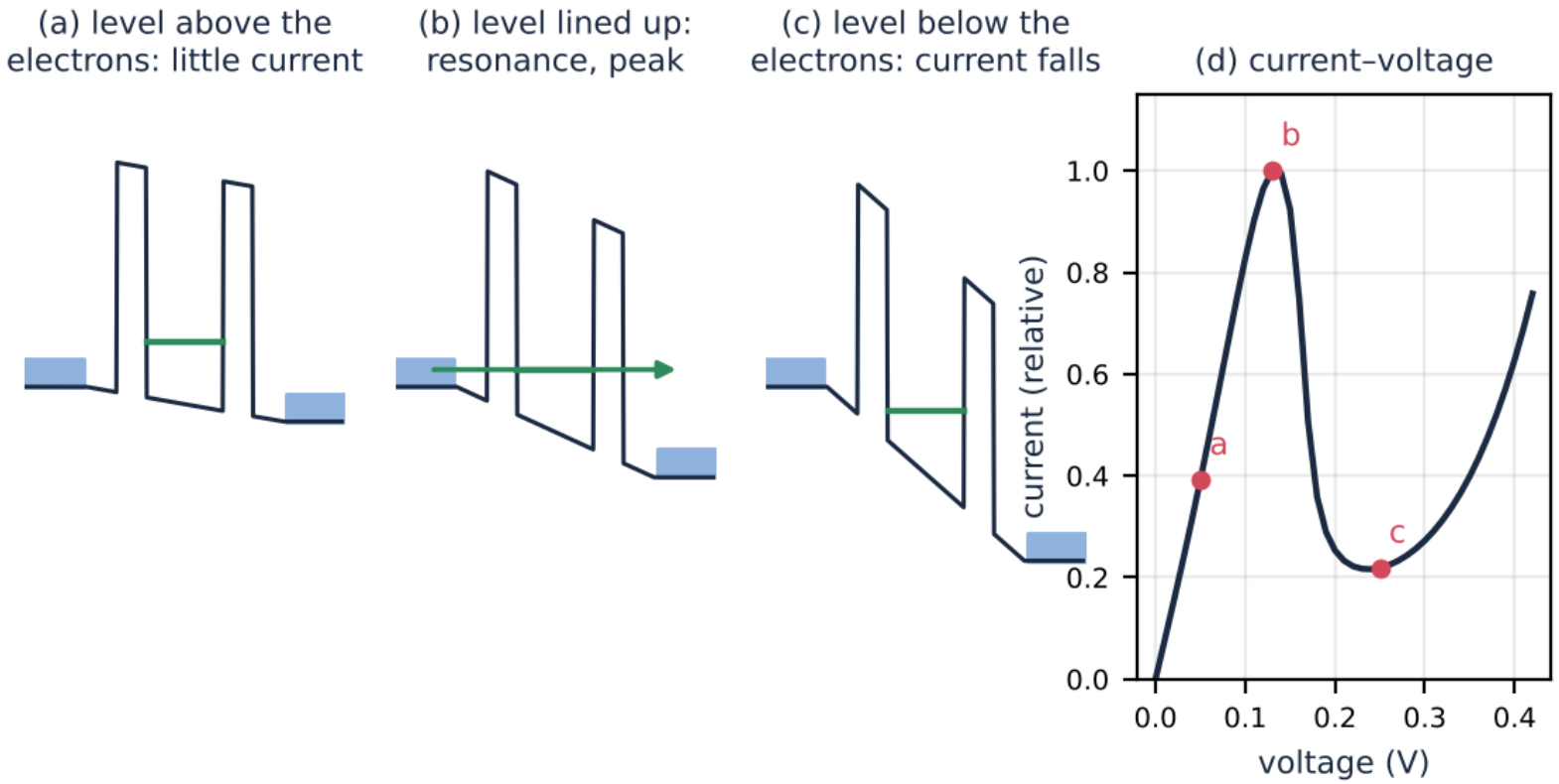


Figure made for this course, calculated at room temperature.

(a) Level above the electrons: little current. **(b)** Level lined up: resonance, **peak**. **(c)** Level below the emitter's band: nothing has the right energy, the current **falls**.

The fastest electronic oscillators

650 GHz

Reddy et al., UC Santa Barbara, 1997

1.98 THz

Asada's group, Tokyo Institute of Technology, 2017

μW

Typical output at the highest frequencies: the main weakness

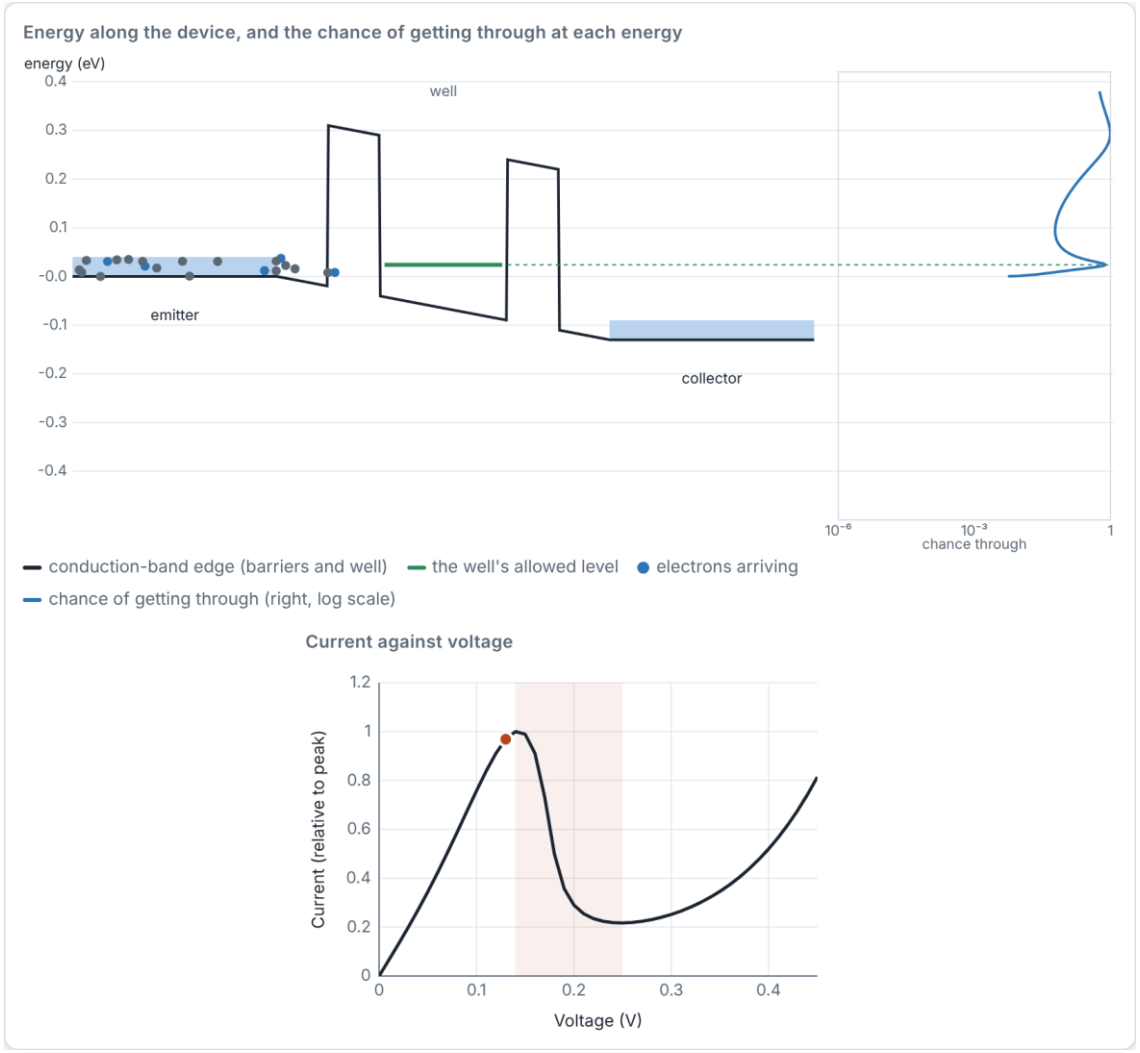
Terahertz waves pass through clothing, paper and plastics: uses in security imaging, gas sensing, inspection and future **6G** short-range links.

Interactive animation 4

Resonant-tunnelling diode

faizverse.github.io/nanoelectronic-devices/L3/resonant-tunnelling-diode.html

- Raise the voltage to the peak: where is the green level?
- Go past the peak: why does the current fall?
- Narrow the well: does the peak move up or down? Why?
- Thicker barriers: what happens to the blue peak?



5

The tunnel transistor

Can a transistor beat 60 mV per decade?

The 60 millivolt problem

Over a barrier

A MOSFET lets electrons over a barrier set by the gate. Only the few 'hot' electrons get over when it is meant to be off.

60 mV per decade

At room temperature the gate must change by at least 60 mV for each tenfold change in current (Lecture 1, Document 2).

Stuck supply voltage

So supply voltages have hardly fallen in twenty years, around 0.7–1 V; power grows with the square of voltage.

Switching by tunnelling

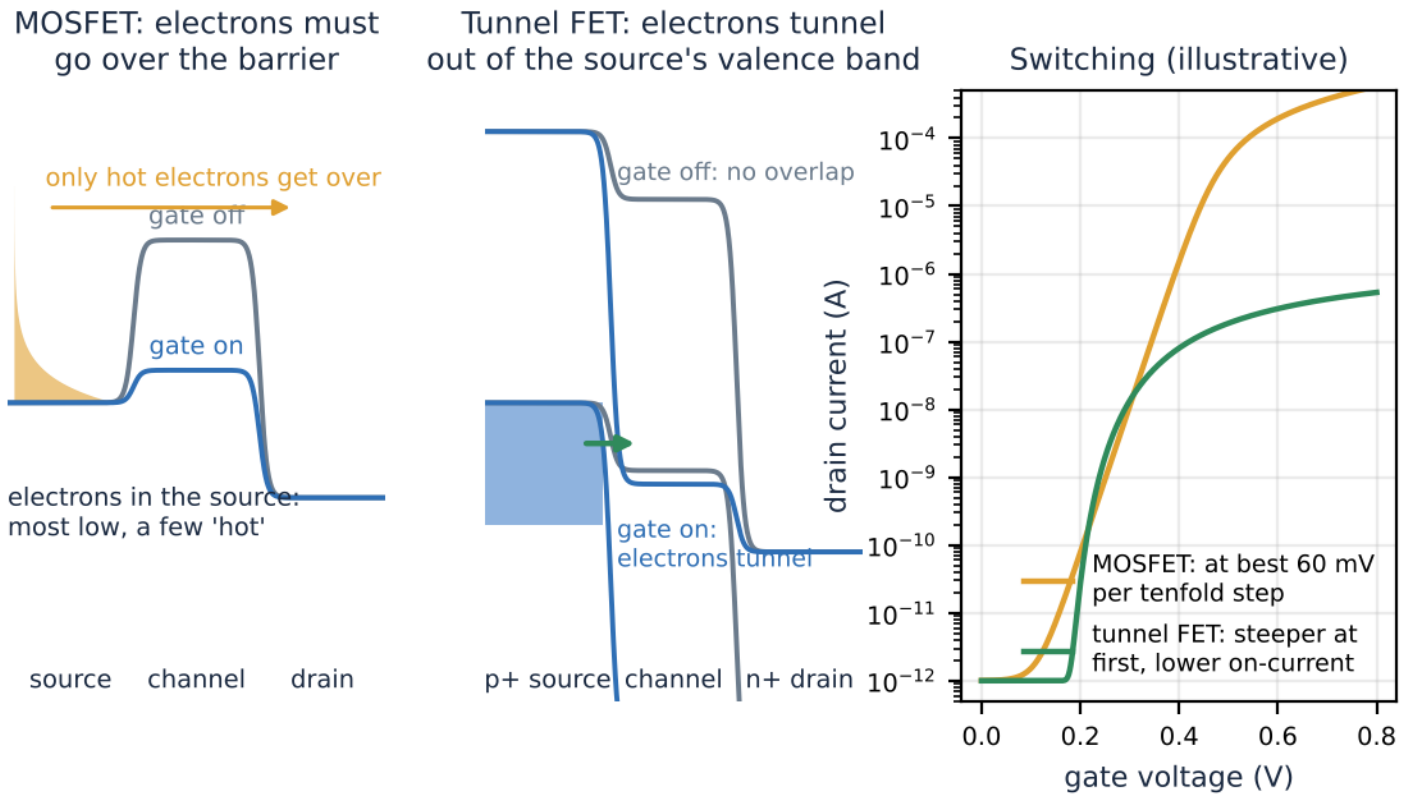


Figure made for this course; schematic.

In a tunnel FET the source is **p-type**. The gate pulls the channel's band below the source's filled valence band, opening a **window**. The window cuts off the hot tail, so it can switch more steeply than 60 mV per decade.

Where tunnel FETs stand

Steeper, but…

Laboratory devices beat 60 mV per decade, usually only over a small range of very low currents.

Too little current

Tunnelling passes far fewer electrons than going over a lowered barrier, so circuits are slow. Defects add unwanted current.

Research, 2026

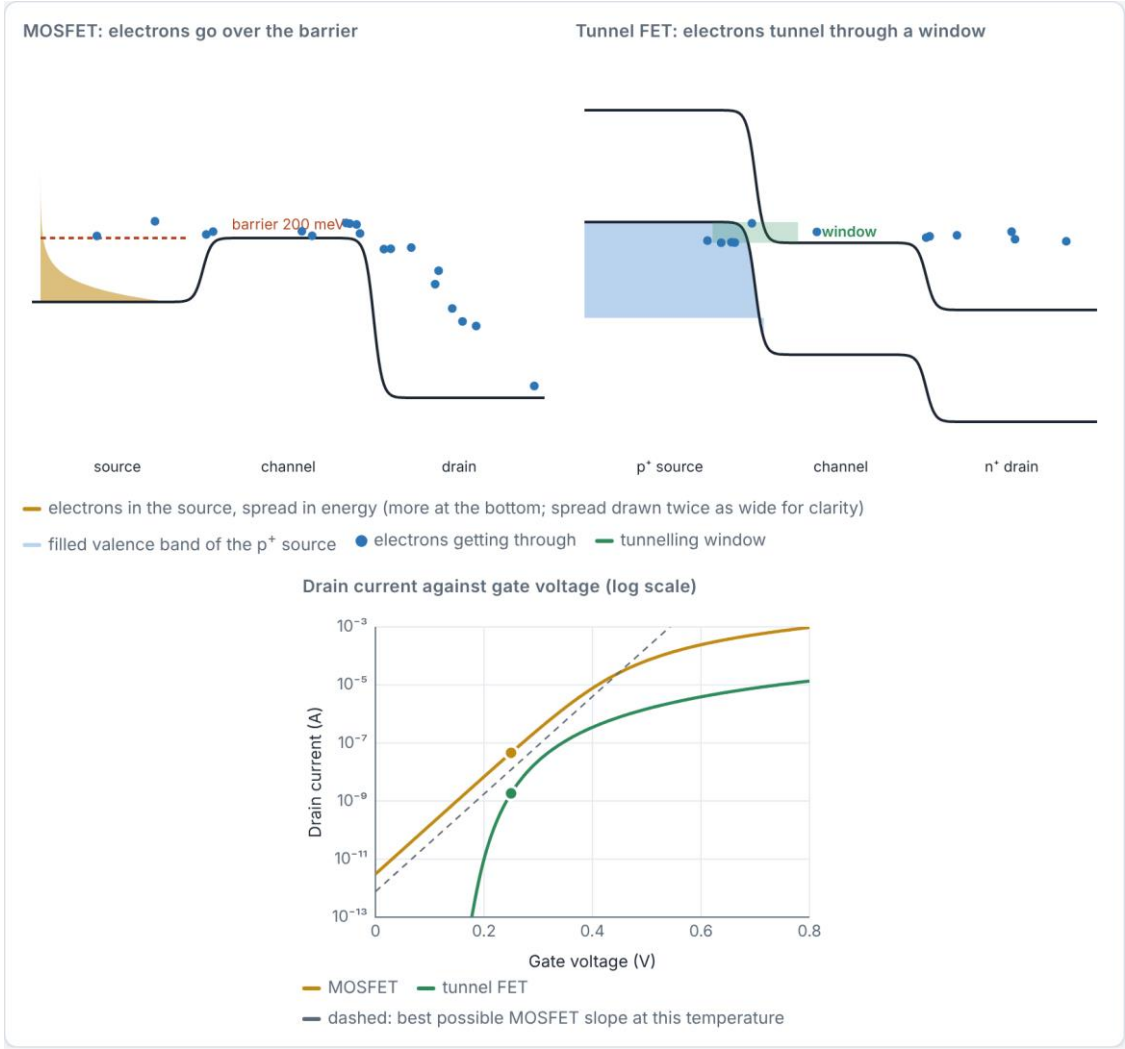
Ge, InAs/GaSb and 2D materials; gate-all-around geometries. No tunnel FET is in mainstream chips yet.

Interactive animation 5

Tunnel FET versus MOSFET

faizverse.github.io/nanoelectronic-devices/L3/tunnel-fet.html

- Sweep the gate from 0 to 0.8 V: where does each turn on?
- Read the steepness at low current: which beats 60 mV per decade?
- Heat both to 400 K: which one gets worse, and why?
- Compare the currents at 0.8 V



6

How these devices are made

All depend on layers or junctions a few nanometres thick.

Making them

Tunnel diodes

1960s: alloy junctions, a dopant-laden pellet melted into a heavily doped wafer. Today: epitaxy, with very heavily doped layers a few tens of nm thick.

RTDs

Molecular beam epitaxy or MOVPE, one atomic layer at a time (Lecture 2, Document 3). Fastest: InGaAs wells, AlAs barriers on InP.

Tunnel FETs

The ordinary transistor process, with source and drain doped opposite types. The hard part: an abrupt, defect-free source junction.

Part 2 in six points

1 Thin junction

Very heavy doping: a ~10 nm wall to tunnel through.

2 Rise and fall

Filled states slide past empty ones: negative resistance.

3 Oscillators

Negative resistance cancels losses; tunnel junctions live on in solar cells.

4 RTD

Two walls and a well: a sharp filter, oscillating up to ~2 THz.

5 60 mV limit

MOSFETs need hot electrons to get over a barrier.

6 Tunnel FET

Steeper switching, but still too little on-current.

Credits

Made for this course

All figures were made for KKKC6353 (scripts in figures-source/). RTD transmission and current are exact calculations. Animation screenshots from faizverse.github.io/nanoelectronic-devices. The tunnel-diode sections draw on slides used in earlier years of the course (third-party, archived).

Sources

Esaki, Phys. Rev. 109, 603 (1958); Chang, Esaki and Tsu, APL 24, 593 (1974); Reddy et al., IEEE EDL 18, 218 (1997); Izumi, Suzuki and Asada, IRMMW-THz (2017); Asada and Suzuki, Sensors 21, 1384 (2021); Ionescu and Riel, Nature 479, 329 (2011). Video: NJPhysics, 'Tunnel diode'.